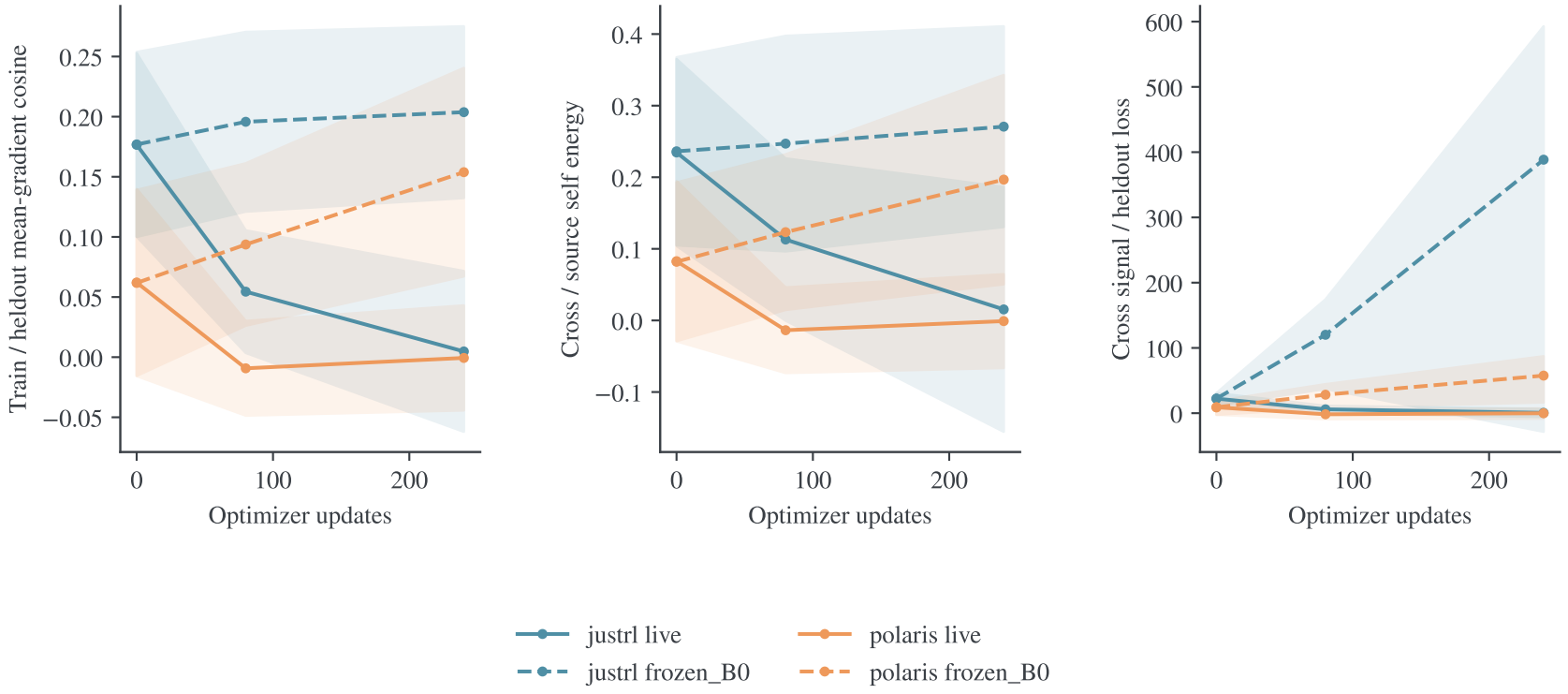
Train / heldout mean-gradient cosine
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
0
100
200
Optimizer updates
Cross / source self energy
0.4
0.3
0.2
0.1
0.0
−0.1
0
100
200
Optimizer updates
Cross signal / heldout loss
600
500
400
300
200
100
0
0
100
200
Optimizer updates
justrl live
justrl frozen_B0
polaris live
polaris frozen_B0